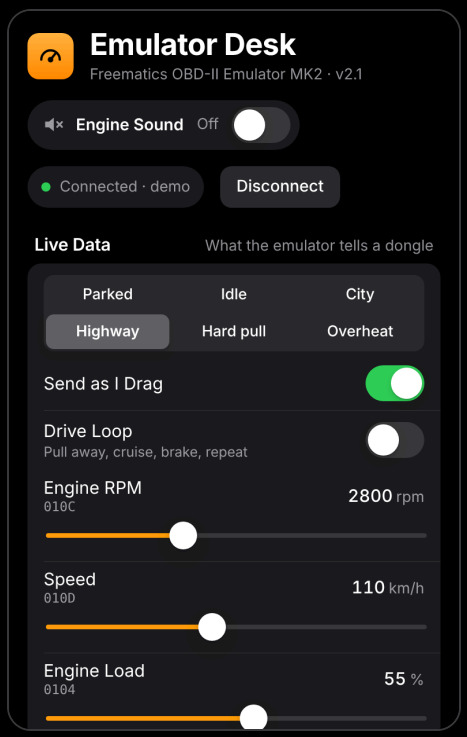
Emulator Desk

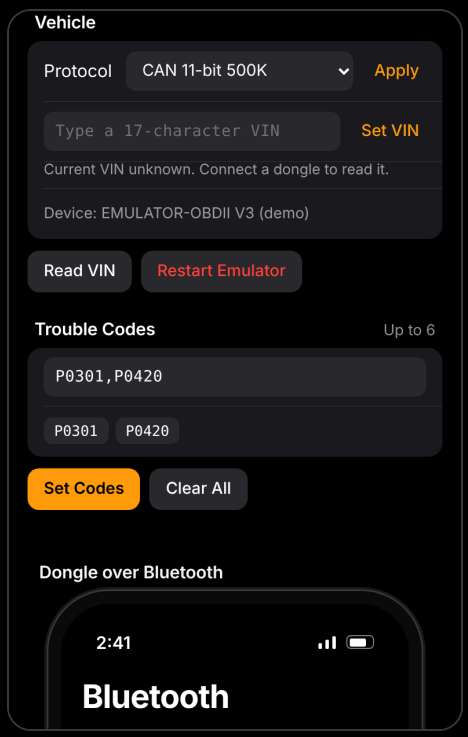
A Mac control panel for the Freematics OBD-II Emulator MK2

App **v2.1**
Document **v2.1**

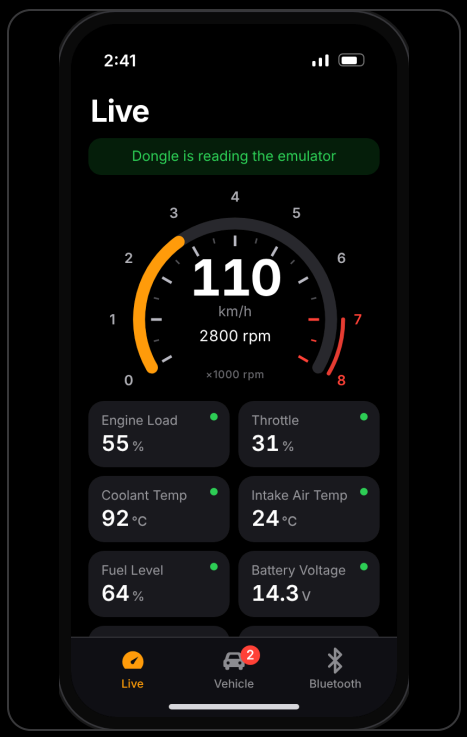
Emulator Desk drives a Freematics MK2 emulator from Google Chrome over its USB cable, so a Mac can do what the Windows-only Freematics GUI does. You set live engine values, the VIN, the bus protocol and trouble codes, and any OBD-II dongle or scan tool plugged into the emulator reads them as if it were in a real car. It runs from a web page, with nothing to install, and follows Apple's Human Interface Guidelines in light and dark mode.



Live data
Ten sliders for the common Mode 01 values, with presets from Parked to Overheat. Values go out as you drag.



Vehicle and codes
A compact panel for protocol, VIN and up to six trouble codes, set or cleared in one tap.



Dongle phone view
A phone with a tab bar. Live has an rpm gauge with redline, Vehicle decodes the VIN, Bluetooth shows the link and what the dongle supports.

10 live PIDs with sliders	**6** driving presets plus a drive loop	**7** bus protocols to choose from	**1** link to open it, nothing to install

- **On your phone**

phone.html is the same Live, Vehicle and Bluetooth app, full screen. Scan the QR code under the phone; on iPhone open it in Bluefy.

- **Dongle over Bluetooth**

Connect a Bluetooth LE ELM327 dongle from the Bluetooth tab. Dots show matches, and Vehicle decodes the VIN it reads into make, model, engine and origin.

- **Good to know**

Sliders read back the emulator's values on connect. Its firmware can't report its VIN, so the VIN box fills in from the dongle.

- **Engine sound**

Flip the Engine Sound switch and a flat-six plays, built from single cylinder firings. It thumps at idle, howls near redline and crackles on lift-off.

Technical setup

Sources, integrations and what runs where

App **v2.1**
Document **v2.1**

Sources and libraries

Piece	Version	What it does	Loaded from	Licence
Emulator Desk	2.1	The whole app: UI, command queue, PID encoding, presets, drive loop, engine sound, dongle check	`index.html` and `phone.html` on GitHub Pages	Written for this app
Freematics AT command set	EMULATOR-OBDII V3 (tested)	Commands the app sends: `ATE0`, `ATZ`, `ATSET`, `ATGET` (PIDs), `ATCLR DTC`	Built into the emulator	Freematics
OBD-II PID formulas	SAE J1979	Byte encodings for each slider (e.g. RPM × 4 as two bytes, temperature + 40)	Hard-coded in the app	Public standard

No third-party libraries, CDNs or web fonts are used; type is the system font.

Integrations

Integration	Used for	Details
Web Serial API	Talking to the emulator	`navigator.serial.requestPort()`, opened at 38400 baud, 8N1. Commands end in `\r`. You pick the port in Chrome's chooser; no auth. Works in Chrome and Edge, not Safari or Firefox.
Freematics OBD-II Emulator MK2	The device being controlled	USB to serial through a CP2102 or CH340 bridge, both with drivers built into macOS. Needs its 12V adapter for power.
Web Bluetooth	Reading a plugged-in dongle (Mac or phone)	`navigator.bluetooth.requestDevice()`, you pick the dongle. Finds a notify and a write characteristic (FFE0, FFF0, 18F0 and similar), sends ELM327 commands: setup (`ATZ`, `ATSP0`), capabilities (`AT@1`, `STI`, `ATDP`, `0100–0160`, `011C`, `0900`) and live polling. Writes in 20-byte chunks, times each reply and keeps a copyable connection log.
NHTSA vPIC	VIN details	`vpic.nhtsa.dot.gov/api/vehicles/DecodeVinValues/{VIN}?format=json`, no key. Sends only the VIN; returns model, engine, body and plant. Maker, country, year and check digit decode offline.
Web Audio API	Engine sound	An AudioWorklet fires six cylinder pulses every two revolutions through exhaust resonators, a muffler filter, EQ and a compressor. Starts only when you flip the switch.
GitHub Pages	Hosting	`davidanthonyalleyne-dot.github.io/emulator-desk`, served from the `main` branch of the public `emulator-desk` repo. Static files only.

What runs where

On your Mac: everything. The page runs in Chrome and sends one command at a time, waiting up to 1.2 seconds for OK or ERROR before the next.

Network: the page load from GitHub Pages, plus one NHTSA lookup per new VIN decoded.

Data kept: nothing. Slider values, VIN and codes live only in the open page and on the emulator itself.

Demo mode: add `?demo` to the address to try the app without hardware. The screenshots in this document use it with sample values.

Regenerating this document

In `docs/overview`, run `node shoot.js` to retake the screenshots in demo mode at 390 × 844, then `node pdf.js` to print `overview.pdf` on Letter pages.